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ALBERTO DINELLI

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WORK EXPERIENCE

2024-09 – Present | **Postdoc**
 Département de biochimie · Université de Genève · *Geneva*
Advisor: Karsten Kruse.

EDUCATION

2021-10 – 2024-08 | **Ph.D. - Physique en Île de France**
 Laboratoire Matière et Systèmes Complexes, Université Paris Cité · *Paris*
Thesis: Scalar Active Matter across scales · **Supervisor:** Julien Tailleur.

2023-10 – 2023-11 · Visiting period at MIT, group of Prof. J. Tailleur · *Cambridge, US*
 2023-07 · Visiting period at YITP, group of Prof. H. Hayakawa · *Kyoto, JP*
 2022-10 · Visiting period at MIT, group of Prof. J. Tailleur · *Cambridge, US*

2019-09 – 2021-07 | **Master in Physics of Complex Systems - International track**
 Politecnico di Torino · *Torino* — Université Paris-Saclay · *Paris*
Final mark: 110/110 cum laude
French M2 mark: 18/20
Thesis: Self-organization of active mixtures · **Supervisor:** Julien Tailleur.

2019-09 – 2020-02 · Visiting period at SISSA, ICTP · *Trieste*

2016-09 – 2019-07 | **Bachelor degree in Physics**
 Università di Pisa · *Pisa*
Final mark: 110/110 cum laude
Thesis: Rayleigh Taylor instability on a thin layer of fluid accelerated by the radiation pressure. · **Supervisor:** Francesco Pegoraro.

TEACHING

2024 - Present · Teaching assistant, *Introduction to computer science* (Bachelor, 1st year) · Université de Genève (*Geneva*)
 2021 - 2024 · Teaching assistant, *Physics of Materials* (Bachelor, 1st year) · IUT Pajol (*Paris*)
 · Teaching assistant, *Photonics* (Bachelor, 2nd year) · IUT Pajol (*Paris*)

PUBLICATIONS

1. [A. Dinelli](#), P. L. Muzzeddu, “Unifying hydrodynamic theory for motility-regulated active matter: from single particles to interacting polymers”. arXiv:2604.09447 (2026)
 2. [A. Dinelli](#), P. L. Muzzeddu, “Multiscale perturbative approach to active matter with motility regulation”. arXiv:2604.09453 (2026)
 3. [A. Dinelli](#), L. Dumoulin, K. Kruse, “Active topological strings in renewing nematopolar fluids”. arXiv:2601.18307 (2026)
 4. Q. M. Nguyen, [A. Dinelli](#), G. Spera, J. Tailleur, “Contact forces in motility-regulated active matter”. arXiv:2507.08964 (In press at Phys. Rev. Lett.).
 5. [A. Dinelli](#), A. Altieri, J. Tailleur, “Random motility regulation as a generic mechanism of community formation”. Phys. Rev. E **114**, 014404 (2026).
 6. T. Lefranc, [A. Dinelli](#), C. Fernández-Rico, R. Dullens, J. Tailleur, D. Bartolo, “Synthetic quorum sensing and absorbing phase transitions in colloidal active matter”. Phys. Rev. X **15**, 031050 (2025).
 7. [A. Dinelli](#), “Scalar active matter across scales”. PhD Thesis (2024).
 8. [A. Dinelli](#), J. O’Byrne, J. Tailleur, “Fluctuating hydrodynamics of active particles interacting via chemotaxis and quorum sensing: static and dynamics”. J. Phys. A **57**, 395002 (2024).
 9. [A. Dinelli](#), J. O’Byrne, A. Curatolo, Y. Zhao, P. Sollich, J. Tailleur, “Non-reciprocity across scales in active mixtures”. Nature Communications **14**, 7035 (2023).
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PRESENTATIONS

Invited talks

- 2025-06 · *Interdisciplinary challenges in the physics of complexity and life 2025 (IntCha25)* · EMBL Heidelberg, Heidelberg
- 2025-05 · *Conservation laws and non-reciprocity* · Universität Münster, Münster

Contributed talks

- 2025-07 · STATPHYS29 · Firenze
- 2023-08 · STATPHYS28 · University of Tokyo, Tokyo
- 2023-08 · *Frontiers in nonequilibrium physics: Active matter, topology and beyond* · YITP, Kyoto
- 2023-06 · EDPIF Scientific Day · Sorbonne University, Paris
- 2022-11 · *Physics of Microbial Motility Workshop* · ESPCI, Paris
- 2022-10 · *Greater Boston Area Statistical Mechanics Meeting* · UMass-Amherst, Amherst
- 2022-06 · CECAM workshop *New Frontiers in Liquid Matter* · Sorbonne Jussieu, Paris

Seminars

- 2026-09 · Laboratoire Jean Perrin, Sorbonne, Paris
- 2026-08 · Work-in-progress seminar · National Institute for Mathematics and Theory in Biology, Chicago
- 2025-10 · Final audition for *thesis prize by Systèmes Complexes 2025* · Maison des sciences de l’homme et de l’environnement, Besançon
- 2024-04 · K. Kruse’s group · Université de Genève, Geneva
- 2024-02 · *Spring College of Physics of Complex Systems 2024* · ICTP, Trieste
- 2024-01 · M. Shelley’s group · CCB, Flatiron Institute, New York
- 2023-10 · D. Nelson’s group · Harvard University, Cambridge
- 2023-10 · P. Mehta and K. Korolov’s groups · Boston University, Boston
- 2023-10 · M. Hagan’s group · Brandeis University, Waltham
- 2023-07 · Advanced Statistical Dynamics Group · YITP, Kyoto

- 2023-03 · Institut Curie Theory Group · Institut Curie, Paris
- 2022-10 · MIT Physics of Living Systems · MIT, Cambridge

Flash talks

- 2026-01 · Journées de Physique Statistique 2026 · Université Paris Cité, Paris
- 2024-01 · Journées de Physique Statistique 2024 · Université Paris Cité, Paris
- 2023-01 · Journées de Physique Statistique 2023 · ENS, Paris
- 2022-01 · Journées de Physique Statistique 2022 · ENS, Paris
- 2021-11 · Multiscale integration in Biological Systems · Institut Curie, Paris

Poster presentations

- 2026-08 · Spin Glass Theory and High-Dimensional Landscapes in Biology · NITMB, Chicago
- 2026-07 · Physics and Life Days · Lyon
- 2026-03 · J'Active Workshop in honour of J. Prost · Les Houches School
- 2023-08 · From Soft Matter to Biophysics 2025 · Les Houches School · **Best Poster Prize**
- 2023-08 · Perspectives on Non-Equilibrium Statistical Mechanics · YITP, Tokyo
- 2023-06 · Active matter at surfaces and in complex environments · MPIPKS, Dresden
- 2023-05 · Workshop on Signatures of Nonequilibrium fluctuations in life · ICTP, Trieste
- 2022-12 · Population dynamics: from rare events to evolution · Université Grenoble Alpes, Grenoble

OTHER INFORMATION

Awards

- 2025 · Best Poster Prize @ Les Houches Winter School
- 2023 · International Mobility Fellowship
- 2021 - 2024 · IdEX International PhD Fellowship
- 2020 - 2021 · Paris-Saclay International Master's Scholarship